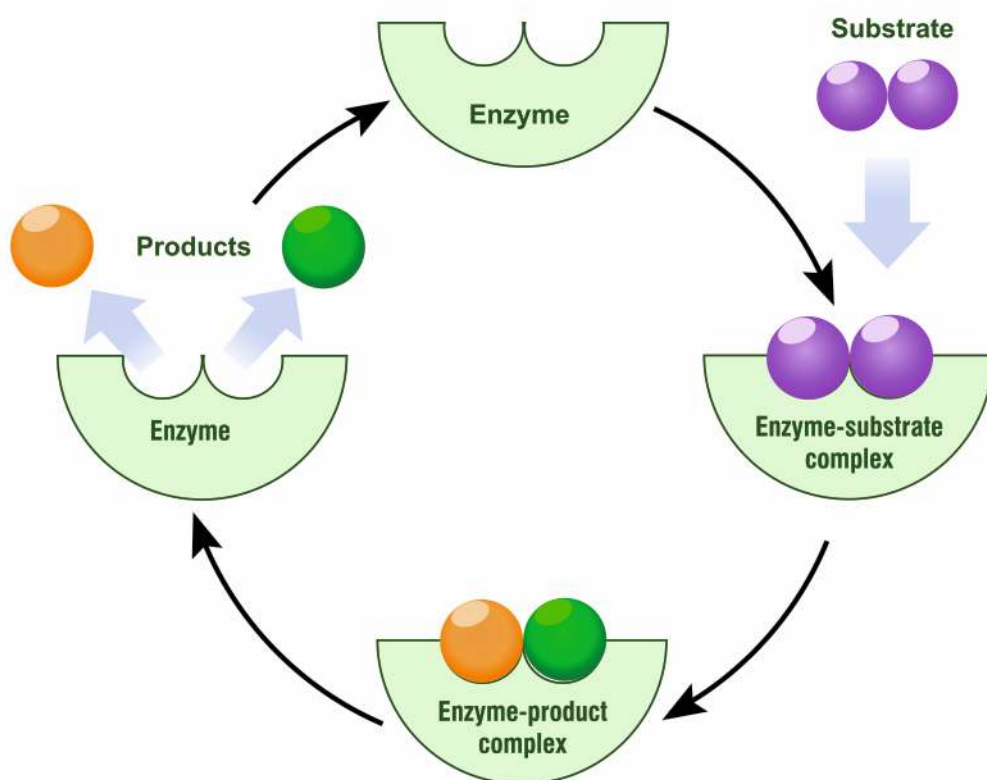
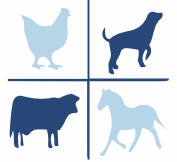




MakZyme- XPL

Premium





Thermostable Extra Strength Xylanase Complex

MakZyme-XPL is a unique enzyme feed additive scientifically developed for consistent & premium performance.

MakZyme XPL Premium is a perfect blend of substrate specific enzymes in powder form to enhance the availability of nutrients especially carbohydrates and protein. MakZyme XPL Premium reduces the gut viscosity, improves nutrient digestibility and releases the hidden nutrients in feed ingredient.

COMPOSITION

- Each Gram Contains
- 1,4 endo xylanase- 1,80,000 BXU/g
- Protease- 40,000 u/g
- Amylase- 2,000 u/g
- Carrier- Starch

1,4 ENDOXYLANASE

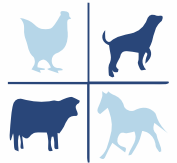
- 1,4 endo-xylanase is used to breakdown xylan, which is a non-starch Polysaccharide found in corn and other grains. Xylanase enzymes hydrolyze glycosidic bonds in xylans. The addition of enzymes to make up to 30% of nutrients more digestible and release of encapsulated nutrients. The additional benefit of xylanase is changing the viscosity due to arabinoxylans and anti-nutritional factors.

Non Starch Polysaccharides Content of Raw Material

Raw Material	Arabino xylans	Beta Glucans	Mannan	Cellulose	Oligosac carides	Others	Total NSP
Corn	4.3	0.3	0.3	2	0.8	0.6	8.3
Wheat	7.1	0.6	0.3	1.8	0.1	0.1	10
Sorghum	3.7	0.1	0.1	1.1	0.2	0.3	5.5
Soybean Meal	0.4	0.7	1.3	9.1	9.6	4.6	25.7
Canola Meal	0.3	0.8	0.5	7.1	3.7	5.6	18
Corn DDGS	11.7	-	0.5	2.7	0.2	10.2	25.3

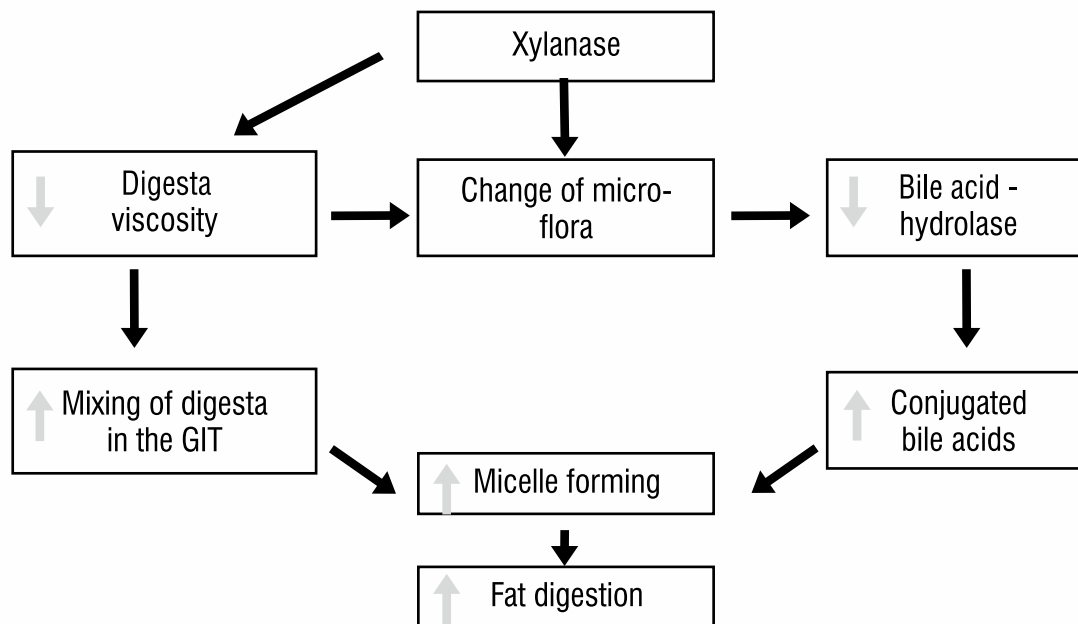
Poultry birds do not produce enzymes for digesting NSP in feed raw materials.

Antinutritional factors (ANF) in NSP cage nutrients & increase the viscosity of digestive tract.



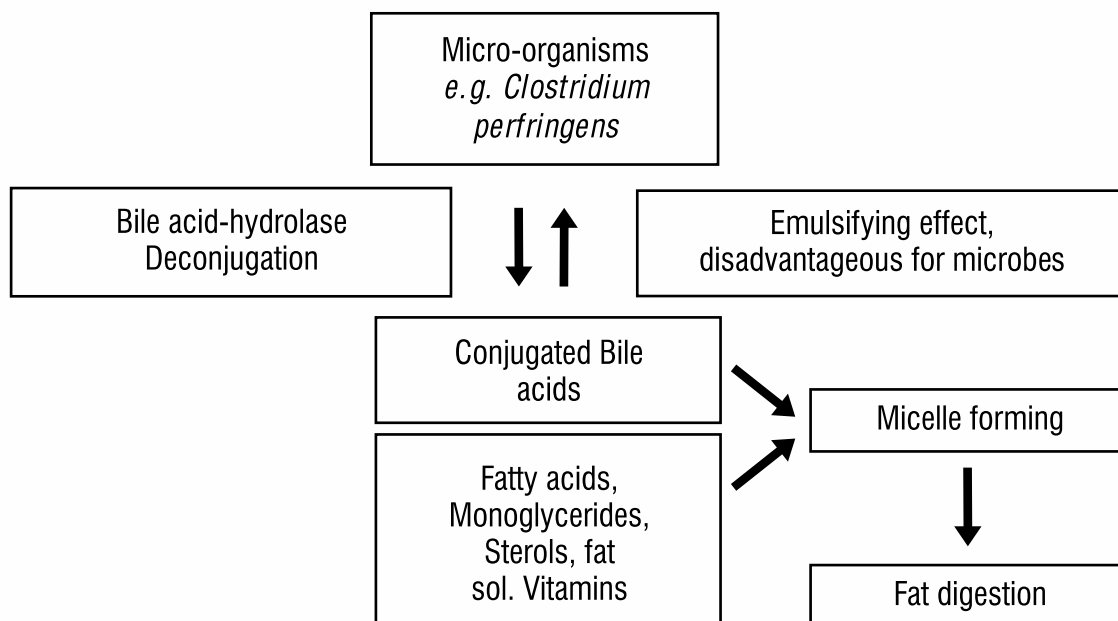
- The interaction between Xylanase supplementation viscosity and fat digestion has been adequately researched. It is known that enzyme supplements, altering substrate availability in the small intestine modify the microflora. Xylanase supplementation reduces the production of bile acid hydrolases by altering the microbe population in the gut. This might result in more conjugated bile acids becoming available and marked the beneficial effect of Xylanase on fat digestibility.

Relation of xylanase and fat digestion (adaptated from HÜBENER *et al.*, 1999)



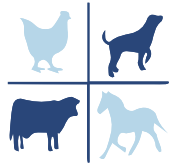
- Long-chain arabinoxylan molecules are breakdown by Xylanase and produce short-chain compounds like Xylo Oligomers which act as a prebiotic. Xylo Oligomer is a nutrient source for beneficial bacteria.

Negative influence of some micro-organisms on fat digestibility. (Cole & Fuller 1984)



MakZyme- XPL

Premium



PROTEASE

Protease is a protein digesting enzyme that breaks down storage proteins binding starch within feed ingredients. This makes the energy from protein bound starch available to the bird to be used for productive purposes. Proteases are also effective in releasing protein anti-nutrients found in ingredients like soybean meal. This function of proteases makes proteins more available. Increases protein digestibility through hydrolysis of storage and structural proteins, and disrupts interactions of proteins with starch and fibre in the diet. Additionally, they target other antinutritional factors in the diet e.g. residual trypsin inhibitors and lectins in soybean meal and some other vegetable proteins, thereby improving nutrient digestibility.

AMYLASE

Acts on starch, increasing its hydrolysis and thereby improving its digestibility by hydrolysis of 1,4 glucosidic linkages. Its actions complement the secretion of endogenous amylase by the bird, freeing up energy to fuel growth. Increasing starch digestibility also reduces the presence of glucose as a potential substrate for non-beneficial bacteria in the latter part of the GIT.

BENEFITS

- Alteration in feed passage rate-MakZyme-XPL Premium reduces the water holding capacity of the gut contents thus increasing the dry matter content stimulating feed intake.
- Reduced viscosity improves nutrient digestion, lower water intake and help to reduce litter problems.
- Addition of MakZyme-XPL Premium also reduces the dry matter outside the body thus has marked impact on excreta volume and composition. Due to protein digestibility lowering of excreta output is observed.
- MakZyme-XPL Premium help to release Nutrients:
 - a) Enzymatic depolymerization renders the NSP free releasing nutrients in the gut and its availability to the animal.
 - b) Starch masked by cell structure is released leading to increase in metabolizable energy.
 - c) Proteins are also released by action of proteases. This release leads to maximum absorption of minerals and results in improvement in nutrient utilization.

FEEDING DIRECTIONS

Broiler/Layers & Breeders: 100 gm/ton of feed
(or) as advised by the Veterinarian

STORAGE

Away from moisture, Hot environment & sunlight

PRESENTATION

25 kg Bag

SHELF - LIFE

24 months from the date of manufacture



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

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